CascadeFilter: Technical Report

This report supplements the main paper, *CascadeFilter: Cost-Efficient Semantic Filtering with Joint Threshold Selection*. References explicitly identified as being in the main paper use its numbering; all other numbered references refer to this report.

Proofs and Supplementary Details

A. Proof of Theorem 2 in the Main Paper

Proof. Fix the entire population, its potential model outputs, and all candidate-specific test rules independently of the sample draw. Write $\theta = N^{-1}\sum_i e_i(\phi)$ and use the filtration and bets in Section I. Uniform ordered sampling gives Eq. (4); in particular its right-hand side is in $[0,1]$ on every history of positive probability.

For $u \le \theta$, the unclipped quantity $(Nu - S_{t-1})/(N-t+1)$ is no larger than $\mu_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$. Clipping to $[0,1]$ preserves $q_t(u) \le \mu_t$. For $q_t < 1$, the admissible bet ensures $1 + \kappa_t(q_t - X_t) \ge 1 - \kappa_t(1-q_t) \ge 1/2$. For $q_t = 1$, any finite nonnegative bet gives a factor at least one; under the null, $\mu_t = 1$ and $X_t = 1$ almost surely, so the factor equals one.

Since κ_t and q_t are measurable before observing X_t,

$$\begin{aligned}\mathbb{E}[M_t^+(u) \mid \mathcal{F}_{t-1}] &= M_{t-1}^+(u)\{1 + \kappa_t(u)(q_t(u) - \mu_t)\} \\ &\le M_{t-1}^+(u).\end{aligned}$$

Thus $M_t^+(u)$ is a nonnegative supermartingale with initial value one. At the fixed sample size, $\mathbb{E}[M_n^+(u)] \le 1$. Markov's inequality at $u = \varepsilon < \theta$ yields $\Pr(M_n^+(\varepsilon) \ge 1/\beta) \le \beta$. This proves the stated fixed-time bound; Ville's inequality also bounds crossing at any time. If the fixed candidate family is itself random independently of the sampling permutation, the same argument holds conditionally on that family, and averaging preserves the bound. □

B. Proof of Lemma 1 in the Main Paper

Proof. The scores $\tilde{z}_i^{(2)}$ are frozen before calibration, so the accept indicator is nested, $A_i^{(2)}(\lambda_{2,m}) \le A_i^{(2)}(\lambda_{2,m'})$ for $m < m'$. With λ_1 fixed, the first term of Eq. (6) in the main paper is unchanged and the second equals $g_i A_i^{(2)}(\lambda_2)$ with the λ_2-free factor $g_i = \mathbf{1}[i \in \mathcal{I}_2^{\max}] D_i^{(1)} \mathbf{1}[\hat{y}_i^{(2)} \ne y_i] \ge 0$, so each e_i is non-decreasing in m. □

C. Proof of Lemma 2 in the Main Paper

Proof. Condition on the fixed population, candidate family, orders, and test rules. On track k, let $m_k^\star = \min\{m : E_{k,m} > \varepsilon\}$, if that set is nonempty. This index is fixed with respect to calibration sampling. If there is no such index, that track cannot certify an infeasible candidate.

Otherwise, certifying any infeasible candidate requires certifying the first infeasible candidate $m_k^\star$: every earlier candidate must pass because the scan stops at its first failure. Therefore

$$\begin{aligned}&\{\text{track } k \text{ certifies an infeasible pair}\} \\ &\quad\subseteq \{M_n^+(\varepsilon; \phi_{k,m_k^\star}) \ge 1/\eta_f\}.\end{aligned}$$

The latter event has probability at most η_f by Theorem 2 in the main paper. Summing over the L_1 tracks gives at most $L_1\eta_f = \delta$. The union bound permits all tracks to reuse the same sample. The argument relies on the fixed order and stopping at the first failed test. □

D. Proof of Theorem 3 in the Main Paper

Proof. Let $\mathcal{G} = \{\forall \phi \in \mathcal{C} : E(\phi) \le \varepsilon\}$, so $\mathbb{P}(\mathcal{G}) \ge 1 - \delta$ by Lemma 2 in the main paper under the theorem assumptions. On $\mathcal{G}$, either $\mathcal{C} \ne \varnothing$ and $\hat{\phi} \in \mathcal{C}$ gives $E(\hat{\phi}) \le \varepsilon$, or the fallback emits y_i on every record and $E(\hat{\phi}) = 0$; either way the target holds, so $\mathbb{P}(\mathrm{Acc}_{\mathcal{M}_L}(\hat{\phi}) < \tau) \le \mathbb{P}(\mathcal{G}^c) \le \delta$. Because $\mathcal{G}$ controls *every* member of $\mathcal{C}$ simultaneously, validity does not depend on which member the label-free cost rule selects, and the $\arg\min$ over the finite $\mathcal{C}$ is attained. □

E. Proof of Theorem 4 in the Main Paper

Proof. There are at most N routing scores at each SLM stage, so sorting them and selecting grid quantiles costs $O(N \log N + G)$ time, including grid materialization. A direct scan evaluates each candidate's routing decisions and sums its cost contributions in $O(N)$ time, giving $O(GN)$ for all candidates. Each candidate's certificate uses at most n constant-time betting updates, and at most G candidates are tested. Early stopping within tracks can reduce this work but does not increase its $O(Gn)$ worst-case bound. Selecting the cheapest certified candidate costs $O(G)$, and producing final routing decisions costs $O(N)$. Cached scalar scores, predictions, and labels occupy $O(N)$ space; candidate costs and certification results occupy $O(G)$. Processing one candidate at a time avoids storing an $N \times G$ error matrix. Uniform ordered sampling can be implemented with a partial permutation in $O(N)$ time and space. The nonempty-grid condition makes $O(N+G)$ additive work dominated by the stated time bound (including $N = 1$). Finally, calibration labels are queried for S, and deployment queries only the uncached members of R, yielding the stated oracle count; for all-oracle fallback take $R = \mathcal{I}$. The analysis counts real-arithmetic operations. An empty grid can instead return the fallback immediately. If the Stage 2 pool is empty, a singleton dummy second-stage grid suffices, since all candidates stop at Stage 1. □

F. Proof of Theorem 1 in the Main Paper

Proof. Routing indicators change only when a threshold crosses a score, proving the piecewise-constant property. For nonconvex cost, take one record with $\tilde{z}^{(1)} = 0.5$, set $c_2 = 1$, and fix λ_2 so Stage 2 accepts. After subtracting C_1, costs at $\lambda_1 = 0.4, 0.8$ are $0, 1$, whereas their midpoint 0.6 has cost $1 > 1/2$, violating convexity.

For a nonconvex feasible set, take one record with both scores 0.5, $a = 0$, $b = 1$, and $\varepsilon = 0.1$. The pair $(0.4, 0.2)$ accepts the correct Stage 1 answer; $(0.8, 0.6)$ uses the oracle. Both have zero error. Their midpoint $(0.6, 0.4)$ accepts the incorrect Stage 2 answer and has error one, so is infeasible.

Increasing λ_1 changes a record's cost from c_{1i} to $c_{1i} + c_{2i} + D_i^{(2)} c_{Li}$; increasing λ_2 can only add oracle cost. Both changes are nonnegative. For fixed λ_1, increasing λ_2 decreases $1 - D_i^{(2)}$, whose coefficient $D_i^{(1)} b_i$ in Eq. (2) of the main paper is nonnegative, so error cannot increase. In contrast, crossing a Stage 1 score changes error by $(1 - D_i^{(2)})b_i - a_i$, which can have either sign. For an explicit example, take two records with Stage 1 scores $0.3, 0.7$, Stage 2 scores both 0.8, $a = (0, 1)$, $b = (1, 0)$, and $\lambda_2 = 0.5$. Their mean errors at $\lambda_1 = 0.2, 0.5, 0.8$ are $1/2, 1, 1/2$, respectively. Thus error need not be monotone in the first threshold. □

G. Evidence Extraction and Stage Details

Table I summarizes the default settings.

TABLE I
DEFAULT CASCADE SETTINGS.

Setting	Value
Small model	Qwen3-4B-Instruct-2507
Embedding model	Qwen3-Embedding-4B
Reranker	Qwen3-Reranker-0.6B
Oracle	GPT-5.5
Oracle thinking mode	Disabled
Local device	One A100-80GB
Evidence window / stride	3 / 1 sentences
RRF constant / extra candidates	60 / 5
Final evidence cap	5 chunks
Stage 1 batch input budget	8,192 tokens
SLM context window	32,768 tokens
Stage 2 / Oracle output limits	16 / 128 tokens
Generation temperature	0
Score temperature T_{temp}	15
Neighbors K	Up to 20, excluding self
NS adjustment weight γ_s	1 at both stages
Log-odds smoothing ε_ℓ	0.001
Shrinkage scale σ_0^2	Median neighborhood mass
Stage 1 / Stage 2 grid sizes	Up to 20 / 32
Score families $\lvert\Psi\rvert$	1
Target τ / failure budget δ	0.9 / 0.05
Betting rule	aGRAPA
Default calibration size	Rounded 1% with floor n_0
Calibration draws per setting	20

1) Evidence locality: For predicates decided by a short passage, the chunk view reduces the input passed to Stage 1.

2) Adaptive Evidence Extraction: Evidence length varies across records. CascadeFilter selects chunks using two score cutoffs. The document d_i is split into sentences (BlingFire) and formed into overlapping three-sentence windows (stride 1). Candidates are ranked by reciprocal rank fusion (RRF) [1] of BM25 and dense (Qwen3-Embedding-4B [2], [3]) scores, $\mathrm{RRF}(c) = \frac{1}{k_0 + \mathrm{rank}_{\mathrm{BM25}}(c)} + \frac{1}{k_0 + \mathrm{rank}_{\mathrm{emb}}(c)}$ (k_0=60), combining exact lexical hits with semantic matches. The first cutoff keeps candidates up to the largest gap in the sorted RRF scores, plus five additional candidates. Qwen3-Reranker-0.6B [2], [4] rescores these candidates. The second cutoff uses the largest gap in reranker scores and keeps at most 5 chunks. Selected chunks are merged in document order, overlapping sentences deduplicated, into $e_{\mathrm{chunk}}(d_i)$.

3) Batched Inference and Base Scoring: CascadeFilter packs multiple query–evidence pairs into one batched context with explicit DocID delimiters,

$$C_B = \mathrm{concat}[\langle \mathrm{DocID}{=}i,\, q_i,\, e_{\mathrm{chunk}}(d_i)\rangle]_{i \in B},$$

where a token budget $B_{\max}$ caps batch size. The SLM emits one binary label per document identifier (DocID) in a single pass. Labels use `Y` for a satisfied predicate and `N` for a negative answer or insufficient evidence. Stage 1 returns one `DocID: Y` or `DocID: N` line per record. For each record, we locate the Y/N label token associated with its unique DocID and extract the alternative-token log probabilities for `Y` and `N` at that position. The prediction and confidence are computed from the same pair of log probabilities. Missing, duplicate, or malformed lines, as well as unavailable alternative log probabilities, cause the affected record to be deferred. Stage 2 applies the same label-token scoring rule to its single-record output. The oracle returns a label on the first line and a short justification on the second.

For Stage 1, define

$$\ell_i^{(1)} = \log p(\mathtt{Y} \mid C_B, \mathrm{prefix}_i) - \log p(\mathtt{N} \mid C_B, \mathrm{prefix}_i).$$

The sign determines $\hat{y}_i^{(1)} = \mathbf{1}[\ell_i^{(1)} > 0]$. We use the absolute log-probability margin to measure confidence in the predicted class, with temperature scaling to obtain comparable routing scores:

$$b_i^{(s)} = \sigma\left(|\ell_i^{(s)}| / T_{\mathrm{temp}}\right), \qquad s \in \{1, 2\}, \quad T_{\mathrm{temp}} = 15.$$

Stage 2 computes $\ell_i^{(2)}$ from the same pair of label-token log probabilities under its full-document prompt and uses the same prediction and confidence rules. This base score is fused with neighborhood evidence into $\tilde{z}_i^{(s)}$ (Section III-C of the main paper), and Stage s accepts when $\tilde{z}_i^{(s)} \geq \lambda_s$. Invalid outputs receive a routing score of $-\infty$, are excluded from score quantiles, and always defer; invalid Stage 1 records are included in $\mathcal{I}_2^{\max}$.

4) Stage 2: Full-Document Filtering: Records that any candidate λ_1 could defer form the maximal Stage-2 subset

$$\mathcal{I}_2^{\max} = \{i \in \mathcal{I} : \tilde{z}_i^{(1)} < \lambda_{1,\max}\}, \qquad \lambda_{1,\max} = \max \Lambda_1, \tag{1}$$

where invalid Stage 1 outputs use the $-\infty$ score convention and $\lambda_{1,\max}$ is the largest first-stage threshold in the grid. The pool therefore contains every record that can reach Stage 2

under a candidate pair. The preparation pass caches predictions and scores for the entire pool. All candidates reuse these results, and Eq. (7) counts the preparation cost once. Stage 2 re-evaluates each record individually, without batch packing, removing the Stage 1 batch error modes (cross-document attention, DocID–label misalignment, evidence truncation), and accepts when $\tilde{z}_i^{(2)} \geq \lambda_2$, with the score recomputed from full-document embeddings.

5) Stage 3: Oracle Fallback: Records rejected by both SLM stages are sent to $\mathcal{M}_L$ with the full document and predicate. The oracle returns a binary label and a short justification. At the calibrated operating points, the Stage 3 rate ranges from 2.5% on CB-Q1 to 31.2% on RV-Q3 (Table III of the main paper). Oracle fallback returns the fixed label $y_i = \mathcal{M}_L(q_i, d_i)$, so its disagreement loss is zero. On CB-Q3, Full SLM has 61.4% benchmark Accuracy (Section V-B of the main paper).

H. Routing-Score Equations

Score construction. Two records can share the same base score $b_i^{(s)}$ while only one deserves acceptance (Figure 1(c) of the main paper), so we want each record's effective threshold to track its difficulty, with no per-record tuning and no held-out labels. A free second cue makes this possible: a record whose embedding neighbors received the same prediction is likelier correct than one that disagrees, so CascadeFilter nudges the base score by how strongly a record's neighbors agree with its prediction.

Neighbor support. Let $v_i^{(s)}$ be the unit-normalized Stage-s embedding and $\mathcal{N}_K^{(s)}(i)$ its top-K cosine neighbors within the stage's own pool ($\mathcal{I}$ at Stage 1, $\mathcal{I}_2^{\max}$ at Stage 2). Both stages embed the predicate together with the record view: extracted evidence at Stage 1 and the full document at Stage 2. Neighbor search uses the stage pool of query–document records without a same-predicate restriction. Thus, Yelp records with different predicates, including records sharing a document, may be neighbors; self-exclusion applies to the query–document record ID. Each record is excluded from its own neighbor set, which contains up to K other records. The weights are $w_{ij}^{(s)} = \max\{\langle v_i^{(s)}, v_j^{(s)} \rangle, 0\}$ and total mass $W_i^{(s)} = \sum_j w_{ij}^{(s)}$. *Neighbor support* $NS_i^{(s)} \in [0,1]$ is the similarity-weighted fraction of those neighbors whose prediction matches i's own $\hat{y}_i^{(s)}$ (set to $\frac{1}{2}$ when $W_i^{(s)} = 0$). It is computed from neighbor predictions.

Shrinkage. Shrinkage moves neighbor support toward the neutral value $\frac{1}{2}$, with weight determined by neighborhood mass:

$$\widetilde{NS}_i^{(s)} = \rho_i^{(s)}\, NS_i^{(s)} + \big(1 - \rho_i^{(s)}\big)\tfrac{1}{2}, \qquad \rho_i^{(s)} = \frac{W_i^{(s)}}{W_i^{(s)} + \sigma_0^2} \in [0,1]. \tag{2}$$

We set σ_0^2 to the median of $W_i^{(s)}$ over the stage pool. This choice keeps $\widetilde{NS}_i^{(s)}$ unchanged when all similarity weights are multiplied by a common positive factor. We set $\rho_i^{(s)} = 0$ when $W_i^{(s)} + \sigma_0^2 = 0$.

Score. The routing score adds a bounded neighbor adjustment to the base score in log-odds space,

$$\tilde{z}_i^{(s)} = \mathrm{lgt}\big(b_i^{(s)}\big) + \gamma_s \left(2\widetilde{NS}_i^{(s)} - 1\right), \tag{3}$$

where $\mathrm{lgt}(b) = \log \frac{b+\varepsilon_\ell}{1-b+\varepsilon_\ell}$ is a smoothed log-odds link (ε_ℓ=10^{-3}) and the neighbor term $2\widetilde{NS}_i^{(s)} - 1 \in [-1,1]$ is positive under agreement, negative under dissent, and zero when uninformative; $\gamma_s \geq 0$ (default 1; γ_s=0 recovers the pure base score) sets its weight.

Routing score variants. The controls in Figure 5(c,d) of the main paper retain the base score $g_i^{(s)} = \mathrm{lgt}(b_i^{(s)})$ and modify the neighborhood term.

NS and shrinkage controls. Removing NS sets $\gamma_1 = \gamma_2 = 0$; removing it from one stage sets only the corresponding γ_s to zero. Removing shrinkage sets $\rho_i^{(s)} = 1$. Constant Shrinkage fixes $\rho_i^{(s)} = 1/2$, so $\widetilde{NS}_i^{(s)} = NS_i^{(s)}/2 + 1/4$.

Similarity weighting. The w/o Similarity Weighting variant assigns unit weights to neighbors and recomputes neighborhood mass and its median. Support then becomes the fraction of matching predictions. If all records have the same nonzero neighbor count, $\rho_i^{(s)} = 1/2$.

Density-only control. The kNN Density-only score is

$$z_{i,\mathrm{density}}^{(s)} = g_i^{(s)} + \gamma_s(2\rho_i^{(s)} - 1).$$

It uses neighborhood mass without prediction agreement; the adjustment is zero when the mass equals its positive median.

Pseudo-label distance support. For Pseudo-label Trust-style, let $d_{i,+}^{(s)}$ and $d_{i,-}^{(s)}$ be the minimum cosine distances within $\mathcal{N}_K^{(s)}(i)$ to neighbors with the same and a different predicted label, respectively. If both groups are nonempty, define

$$T_i^{(s)} = \frac{d_{i,-}^{(s)} + \eta}{d_{i,+}^{(s)} + d_{i,-}^{(s)} + 2\eta}, \qquad \eta = 10^{-6}.$$

Otherwise, set $T_i^{(s)} = 1/2$. This variant substitutes $T_i^{(s)}$ for $NS_i^{(s)}$ while retaining the original shrinkage weights.

Randomized neighbor labels. Randomized Neighbor Labels permutes the neighbor predictions within each stage pool, preserving the class counts. Agreement compares the permuted neighbor labels with each record's original prediction. Base scores, output predictions, embeddings, similarity weights, and shrinkage are unchanged. The permutation is fixed before calibration and reused across draws. Each scoring variant recomputes its affected routing scores, maximal Stage 2 pool, and threshold grids before calibration, with one fixed scoring configuration per run.

Fixed routing scores. Neighbor predictions and weights are fixed independently of the calibration draw. The resulting $\widetilde{NS}_i^{(s)}$ and $\tilde{z}_i^{(s)}$ therefore remain fixed as candidate thresholds vary. Each stage applies $A_i^{(s)}(\lambda_s) = \mathbf{1}[\tilde{z}_i^{(s)} \geq \lambda_s]$, giving nested acceptance sets as required in Section IV of the main paper.

Threshold grids. We take quantiles of valid routing scores at equally spaced levels from 0 to 1, including both endpoints:

up to 20 levels at Stage 1 and 32 at Stage 2. Duplicate quantile values are merged. L_1 is the number of distinct Stage 1 thresholds, and the test level on each track is δ/L_1. Both grids are fixed before drawing S.

I. Single-Candidate Betting Test

Condition on a fixed finite population and fixed potential model outputs. Decoding, batching, retrieval, score construction, candidate grids, and test rules are fixed independently of the calibration draw. The oracle target is a fixed function or a set of cached potential labels generated independently of that draw.

Let $\pi_1, \ldots, \pi_n$ be the first n entries of a uniform random permutation of $\mathcal{I}$, with fixed $1 \le n \le N$. Use this draw order or an independent uniform reshuffling of the selected sample. For a fixed candidate define $X_t = e_{\pi_t}(\phi)$, $S_{t-1} = \sum_{k<t} X_k$, and $\mathcal{F}_t = \sigma(\pi_1, X_1, \ldots, \pi_t, X_t)$, including the fixed background information. Since the next index is uniform among the remaining indices,

$$\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = \frac{N\theta - S_{t-1}}{N - t + 1}, \qquad \theta = \frac{1}{N}\sum_i e_i(\phi). \quad (4)$$

For a target $u \in (0,1)$, set $q_t(u) = \mathrm{clip}_{[0,1]}\{(Nu - S_{t-1})/(N-t+1)\}$ and

$$M_t^+(u) = \prod_{k=1}^{t}[1 + \kappa_k(u)(q_k(u) - X_k)], \quad M_0^+(u) = 1. \quad (5)$$

We use approximate growth-rate adaptive (aGRAPA) betting [5], with $m_t = (1/2 + S_{t-1})/t$, $v_t = (1/4 + \sum_{j<t}(X_j - m_j)^2)/t$, and raw bet $\widetilde{\kappa}_t(u) = \max\{0, (q_t(u) - m_t)/[v_t + (q_t(u) - m_t)^2]\}$. We set $\kappa_t(u) = \min\{\widetilde{\kappa}_t(u), 1/[2(1-q_t(u))]\}$ for $q_t(u) < 1$, and $\kappa_t(u) = \widetilde{\kappa}_t(u)$ for $q_t(u) = 1$. These bets are finite, nonnegative, and $\mathcal{F}_{t-1}$-measurable, and satisfy $\kappa_t(u)(1 - q_t(u)) \le 1/2$.

Certify a candidate at level β if $M_n^+(\varepsilon) \ge 1/\beta$. This rejects the excessive-error null at the fixed sample size. All proof statements are conditional on the fixed setup and therefore remain valid after integrating over independent preprocessing randomness.

1) Zero-error calibration size: We set

$$n_0 = \min\{1 \le m \le N : M_m^+(\varepsilon) \ge L_1/\delta\},$$

with the wealth evaluated on an all-zero loss sequence using the aGRAPA rule specified above. This calculation is performed before drawing S. If the threshold is not reached within N steps, we set $n_0 = N$ and use the full population for calibration.

For an all-zero sample with $q_t(u) < 1$ for every $t \le n$, the admissible cap $\kappa_t(u) \le 1/[2(1 - q_t(u))]$ gives

$$U_n(u) = \prod_{t=1}^{n}\left(1 + \frac{q_t(u)}{2(1 - q_t(u))}\right), \qquad M_n^+(u) \le U_n(u). \quad (6)$$

2) Calibration and evaluation: We draw S uniformly without replacement from the full population $\mathcal{I}$ and retain its random draw order. Each track tests Stage 2 thresholds from strict to lenient at level δ/L_1, stopping at the first failed test. Selection returns the cheapest certified pair; an empty certified set uses oracle fallback for the whole population. We repeat calibration over 20 independent draws per setting and report means and sample standard deviations for benchmark Accuracy and oracle agreement. A run counts as a violation when the selected cascade's agreement with the fixed oracle over the full population is below τ. Benchmark Accuracy and Macro-F1 are evaluated against the reference labels f^*.

The repeated-draw evaluation reuses a fixed full-population oracle-label map together with the prepared SLM outputs. For each draw, threshold fitting receives only the labels indexed by that draw's sample S; labels outside S are used for fallback or final evaluation. The full map supplies the reference for population-wide oracle agreement, while benchmark labels supply the reference for Accuracy and Macro-F1. Each simulated deployment is charged for its own $S \cup R(\phi)$, even when labels are replayed from a shared offline cache. Constructing the full oracle reference is an offline evaluation expense, separate from the calibration and fallback cost of one deployment.

3) Calibration-budget options: The default budget uses rounded one percent of N with the zero-loss floor n_0. For a nominal ratio r, replacing 0.01 by r in that rule gives $n_{\mathrm{floor}}(r) = \min\{N, \max\{\mathrm{ROUND_HALF_UP}(rN), n_0\}\}$. A fixed budget supplied before sampling instead sets n directly. Its effective ratio is n/N. The default floor depends on N, τ, δ, and the actual track count L_1; a change in these inputs changes the corresponding floor calculation. Model outputs, score rules, and candidate grids remain fixed when changing only certification targets or sample budgets.

J. Extensions of the Certification Target

Remark 1 (Other losses and conditional metrics)**.** *The single-candidate proof applies to a fixed population of losses $\ell_i(\phi) \in [0,1]$ through the same conditional-mean identity and nonnegative betting factors. For a fixed finite family and fixed track orders, the first-error argument controls false certification of mean-loss constraints. An end-to-end guarantee requires a fallback that satisfies the target, a separately certified fallback with an allocated failure budget, or an option to return no certified decision. Oracle fallback has zero loss for disagreement with the same fixed oracle.*

Precision and recall use different conditional target populations. For a subgroup of known size fixed independently of calibration sampling, conditioning on its sample count gives a uniform sample without replacement within that subgroup. The same test applies using the subgroup size and its random-order sample, provided at least one member is observed. An empty target group requires an explicit metric convention.

Recall requires information about the size of the positive-label population. For precision, the predicted-positive population varies with the candidate and can depend on oracle

fallback labels. Certification of these ratios requires the relevant population information, a valid conditional-population or ratio procedure, and an allocation of the failure budget across the resulting tests.

K. Cost Accounting with Cached Predictions

Search overhead and prediction reuse. Equation (1) in the main paper defines the execution cost of a selected plan. Preparation evaluates Stage 1 on all records and Stage 2 on $\mathcal{I}_2^{\max}$, then caches these predictions. With residual set $R(\phi) = \{i : D_i^{(1)} D_i^{(2)} = 1\}$, the total cost for the prepared batch is

$$C_{\text{batch}}(\phi; S) = C_{\text{prep}} + C_1 + \sum_{i \in \mathcal{I}_2^{\max}} c_{2i} + \sum_{i \in S \cup R(\phi)} c_{Li}, \quad (7)$$

where C_{prep} covers retrieval, embedding, BM25, reranking, indexing, and threshold-search computation. The Stage 2 term covers every record in $\mathcal{I}_2^{\max}$, including records that the selected plan accepts at Stage 1. Each model invocation is charged once. Threshold selection minimizes $C(\phi)$, whereas prepared-batch reporting uses $C_{\text{batch}}(\phi; S)$. These objectives can select different threshold pairs. The certification argument also permits minimizing C_{batch} over the certified set, which changes the cost objective.

Token and local-compute charges. Shared Stage 1 prompt overhead and generated tokens are counted once per batch and allocated across its records. Oracle input and output lengths include the complete prompt and label-plus-justification response. Nominal input/output token rates are \$0.2/\$0.2 for the SLM and \$5/\$30 for the oracle, in USD per million tokens. The same nominal rates are used across methods. When local-compute charges are included in C_{prep}, they are computed from measured resource usage and its nominal unit rate, with matching units (e.g., device-hours and USD per device-hour). Preparation is charged once per run; query-specific proxy training or cascade assembly belongs to the corresponding baseline's setup.

The oracle labels in S are reused for fallback, giving $|S \cup R(\phi)| = n + |R(\phi)| - |S \cap R(\phi)|$ oracle evaluations. Equivalently, the oracle term charges S and $R(\phi) \setminus S$, without double counting calibration calls. Dividing $C_{\text{batch}}(\phi; S)$ by N gives nominal cost per query–document pair. The reported unweighted average gives equal weight to the 13 setting-level costs. Offline construction of the full-population oracle evaluation map is accounted for separately in Section I2.

L. Threshold Selection Example

The example in Section III of the main paper uses fixed scores and illustrative stage costs. If the short evidence gives an incorrect prediction and the full document gives a correct one, the second pair corrects the answer for three additional cost units. If both predictions disagree with the oracle, oracle fallback returns the reference label. A record can also have a correct prediction from short evidence and an incorrect one from the full document.

TABLE II
PRIMARY NOTATION.

Symbol	Meaning
$\mathcal{I}$	Fixed deployment population.
N	Population size, $\lvert\mathcal{I}\rvert$.
S	Calibration sample drawn without replacement.
n	Calibration sample size, $\lvert S\rvert$.
$\mathcal{M}_S$	Small language model.
$\mathcal{M}_L$	Reference oracle model.
τ	Target oracle-relative accuracy.
ε	Maximum error rate, $1 - \tau$.
δ	Total certification failure budget.
y_i	Oracle label of record i.
$\hat{y}_i^{(s)}$	Fixed SLM prediction at Stage s.
$\tilde{z}_i^{(s)}$	Adaptive routing score at Stage s.
$A_i^{(s)}$	Stage-s acceptance indicator.
$D_i^{(s)}$	Stage-s deferral indicator.
Λ_1	Stage-1 threshold grid.
Λ_2	Stage-2 threshold grid.
Φ	Candidate grid, $\Lambda_1 \times \Lambda_2$.
L_1	Number of Stage-1 thresholds.
L_2	Number of Stage-2 thresholds.
η_f	Per-track test level, δ / L_1.
ϕ	Threshold pair (λ_1, λ_2).
$\mathcal{I}_2^{\max}$	Maximal Stage-2 population.
c_{si}	Processing cost of record i at stage s.
$C(\phi)$	Selected-plan execution cost.
$R(\phi)$	Oracle fallback set.
$C_{\text{batch}}(\phi; S)$	Prepared-batch execution cost.
$e_i(\phi)$	Oracle-relative error of record i.
$E(\phi)$	Population mean oracle-relative error.
$r_3(\phi)$	Stage-3 fallback rate.
M_n^+	Without-replacement betting wealth.
$\mathcal{C}$	Set of certified threshold pairs.
$\hat{\phi}$	Least-cost certified threshold pair.
n_0	Sample-size floor from zero-loss betting.

A threshold pair applies to the entire population. Calibration tests its average disagreement with the oracle, and selection returns the cheapest certified pair. The example table uses the plan cost in Eq. (1) of the main paper; Eq. (7) accounts for execution with cached predictions.

M. Notation

N. Baseline Variants and Shared Settings

The experiments use the accuracy-oriented A variant of BARGAIN and the accuracy-guaranteed +G variant of Task Cascades. These variants are denoted BARGAIN and Task Cascades, respectively, in the main text and result tables.

Target-based baselines use the common accuracy target $\tau = 0.9$; methods with a statistical failure parameter use $\delta = 0.05$. The comparisons share the SLM and oracle models, nominal token rates, and per-setting oracle-label budget n in Section V-A of the main paper. Each method retains its own routing or selection procedure: direct prediction for Full SLM, proxy routing for LOTUS and BARGAIN, cluster-based label inference for UniVote/SimVote, a query-specific trained proxy for ScaleDoc, and task assembly followed by accuracy certification for Task Cascades. Full SLM has no threshold-calibration stage. Accuracy and Macro-F1 use the

same retained benchmark populations and reference labels across methods.

Closest baseline procedures.

BARGAIN uses its accuracy-oriented A variant [6]. Oracle-labeled records calibrate the agreement of accepted proxy predictions. The sampling and threshold-calibration procedure follows the baseline's default configuration, with the accuracy and failure gates aligned to $\tau = 0.9$ and $\delta = 0.05$.

ScaleDoc [7] uses precomputed document representations and query-specific contrastive proxy training, followed by calibration of lower and upper acceptance thresholds from oracle-labeled score distributions. Proxy training follows the baseline's default configuration, with the accuracy target aligned to $\tau = 0.9$.

Task Cascades uses its accuracy-guaranteed +G variant [8]. An LLM agent proposes surrogate operations; candidate tasks combine these operations with document portions; greedy assembly selects cost-reducing tasks on development data; and a separate validation step adjusts the assembled cascade's thresholds along a prescribed sequence of score-based offsets. Candidate generation, assembly, and the development/validation split follow the baseline's default configuration, with the final accuracy-certification gate aligned to $\tau = 0.9$ and $\delta = 0.05$.

Cost scope. End-to-end nominal accounting includes each method's preparation, proxy training or cascade assembly where applicable, calibration calls, and deployed inference. A cached model invocation is charged once within a run. Oracle labels used for calibration are reused when their records later require fallback. The offline oracle reference used only for full-population evaluation is treated separately as described in Sections I2 and K.

O. Dataset Construction and Reference Annotations

A record is a query–document pair, so the same text evaluated against different predicates contributes distinct records. Tables I and III of the main paper give predicate definitions, source citations, and retained workload sizes. For non-Yelp tasks, predicate definitions follow the cited workload; Codebase and RV-Q2/Q3 use its model-generated reference labels.

The final Yelp population contains 100K pairs spanning 50 manually written aspect predicates and 2,000 cleaned texts. A pair's consensus reference label is retained when Claude Sonnet 4.6, ChatGLM 5.2, and DeepSeek-V4-Pro agree. The reported human comparison uses the 1K audit sample and measures agreement with these reference annotations. The 100K count refers to the post-consensus evaluation population. Benchmark reference labels f^* are used for Accuracy and Macro-F1, while the fixed oracle labels y_i are used for calibration and oracle-agreement evaluation.

P. Joint-Search Ablation Protocol

The *w/o Joint Search* variant selects the two thresholds sequentially. It uses the same population, calibration sample S, frozen grid Φ, cost function $C(\phi)$, accuracy target τ, and failure budget δ as the full method. The certification procedure is unchanged, producing the same certified set $\mathcal{C}$.

First, we fix the Stage 2 threshold to the most stringent grid value $\lambda_2^{\mathrm{ref}} = \max \Lambda_2$ and select the minimum-cost certified Stage 1 threshold. We then fix this Stage 1 threshold and select the minimum-cost certified Stage 2 threshold along its track. Both selections evaluate the complete cascade cost $C(\lambda_1, \lambda_2)$.

Sequential threshold selection:

Require: Certified set $\mathcal{C}$, grids Λ_1, Λ_2, cost $C(\cdot)$
1: **if** $\mathcal{C} = \emptyset$ **then**
2: **return** all-oracle fallback
3: **end if**
4: $\lambda_2^{\mathrm{ref}} \leftarrow \max \Lambda_2$
5: $\lambda_1^\star \leftarrow \arg\min_{\lambda_1:(\lambda_1,\lambda_2^{\mathrm{ref}})\in\mathcal{C}} C(\lambda_1, \lambda_2^{\mathrm{ref}})$
6: $\lambda_2^\star \leftarrow \arg\min_{\lambda_2:(\lambda_1^\star,\lambda_2)\in\mathcal{C}} C(\lambda_1^\star, \lambda_2)$
7: **return** $(\lambda_1^\star, \lambda_2^\star)$

The full method minimizes cost jointly over $\mathcal{C}$, whereas this variant selects the first-stage threshold before optimizing the second. Both methods retain the same cascade-level certification and oracle-relative accuracy guarantee.

REFERENCES

[1] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, "Reciprocal rank fusion outperforms condorcet and individual rank learning methods," in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, ser. SIGIR '09. New York, NY, USA: Association for Computing Machinery, 2009, pp. 758–759. [Online]. Available: https://doi.org/10.1145/1571941.1572114

[2] Y. Zhang, M. Li, D. Long, X. Zhang, H. Lin, B. Yang, P. Xie, A. Yang, D. Liu, J. Lin, F. Huang, and J. Zhou, "Qwen3 embedding: Advancing text embedding and reranking through foundation models," 2025. [Online]. Available: https://arxiv.org/abs/2506.05176

[3] Qwen Team, "Qwen3-Embedding-4B," Hugging Face model card, 2025, accessed: 2026-07-02. [Online]. Available: https://huggingface.co/Qwen/Qwen3-Embedding-4B

[4] ——, "Qwen3-Reranker-0.6B," Hugging Face model card, 2025, accessed: 2026-07-02. [Online]. Available: https://huggingface.co/Qwen/Qwen3-Reranker-0.6B

[5] I. Waudby-Smith and A. Ramdas, "Estimating means of bounded random variables by betting," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 86, no. 1, pp. 1–27, Feb. 2024. [Online]. Available: https://doi.org/10.1093/jrsssb/qkad009

[6] S. Zeighami, S. Shankar, and A. Parameswaran, "Cut costs, not accuracy: LLM-powered data processing with guarantees," *Proc. ACM Manag. Data*, vol. 3, no. 6, Dec. 2025. [Online]. Available: https://doi.org/10.1145/3769776

[7] H. Zhang, Y. Hui, Y. Liu, and H. Zhang, "ScaleDoc: Scaling LLM-based predicates over large document collections," *Proc. ACM Manag. Data*, vol. 4, no. 3, May 2026. [Online]. Available: https://doi.org/10.1145/3802106

[8] S. Shankar, S. Zeighami, and A. Parameswaran, "Task Cascades for efficient unstructured data processing," *Proc. ACM Manag. Data*, vol. 4, no. 1, Apr. 2026. [Online]. Available: https://doi.org/10.1145/3786702